Linear Regression

Part -2

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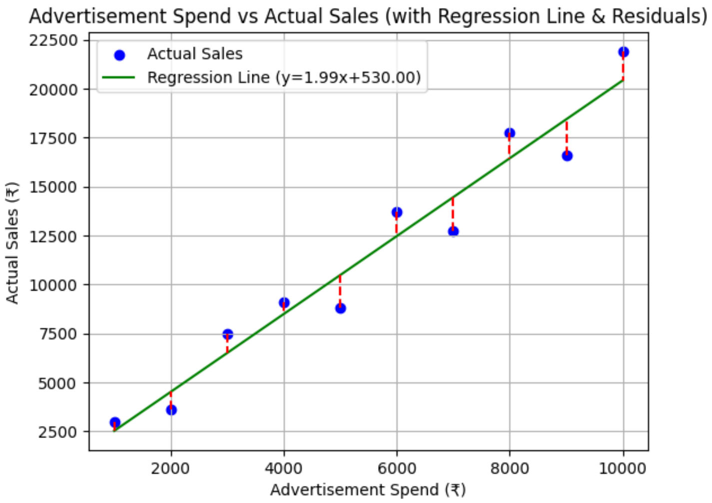
Closed form solution for 2-D case



Error Function (Mean Squared Error):

$$\text{Error} = \frac{1}{n}\sum_{i=1}^{n}(mx_i + c - y_i)^2$$

Set partial derivatives to zero

$$\frac{\partial E}{\partial m} = \frac{2}{n}\sum_{i=1}^{n} x_i(mx_i + c - y_i) = 0, \qquad \frac{\partial E}{\partial c} = \frac{2}{n}\sum_{i=1}^{n}(mx_i + c - y_i) = 0$$

This gives the normal equations:

$$m\sum x_i^2 + c\sum x_i = \sum x_i y_i$$

$$m\sum x_i + nc = \sum y_i.$$

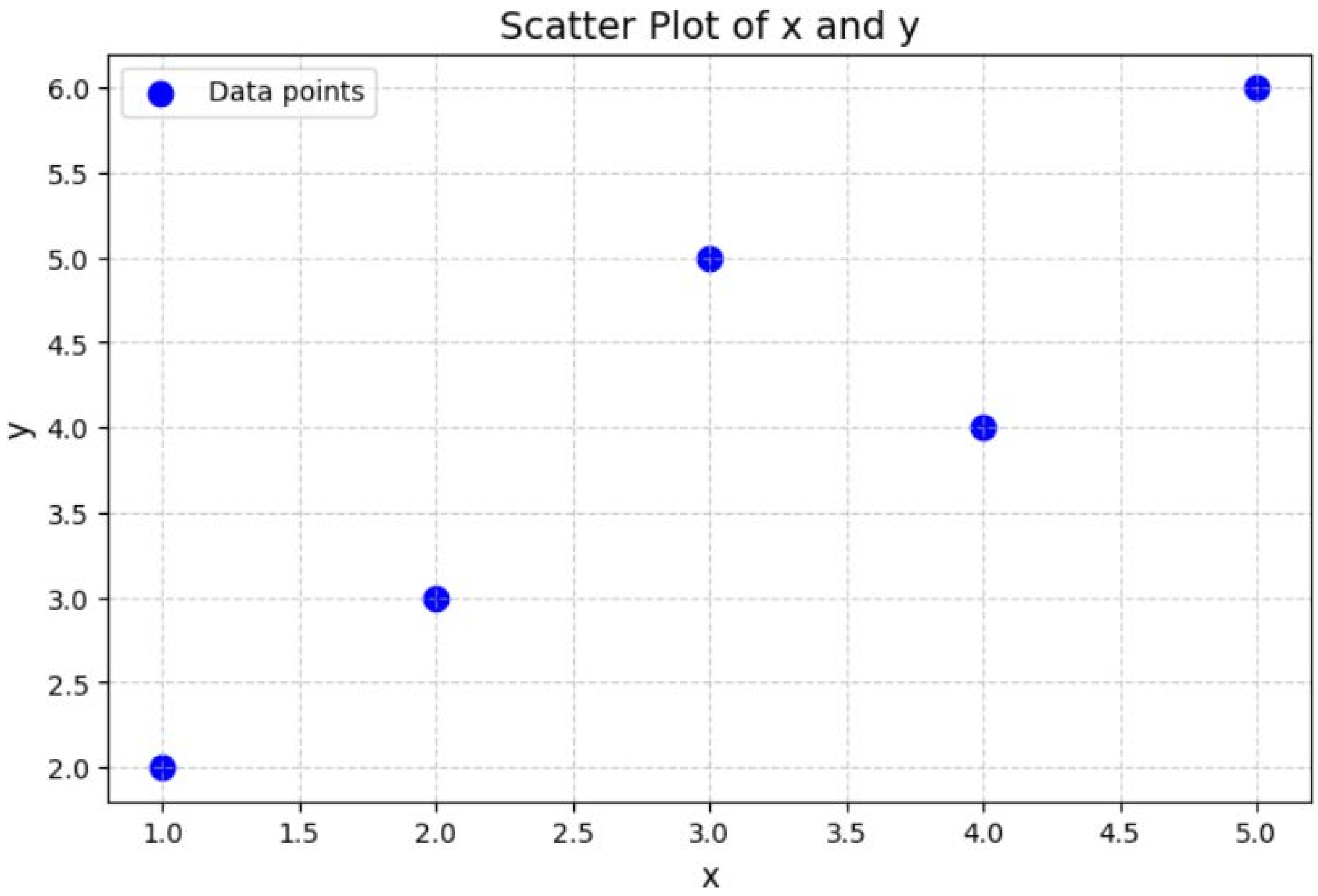
Solve the 2×2 linear system

Slope ***m*** and intercept ***c*** that minimize the Mean Squared Error

$$m = \frac{n \times \sum x_i y_i - (\sum x_i) \times (\sum y_i)}{n \times \sum x_i^2 - (\sum x_i)^2}, \quad c = \frac{\sum y_i - m \times \sum x_i}{n}$$

Example

x	y
1	2
2	3
3	5
4	4
5	6

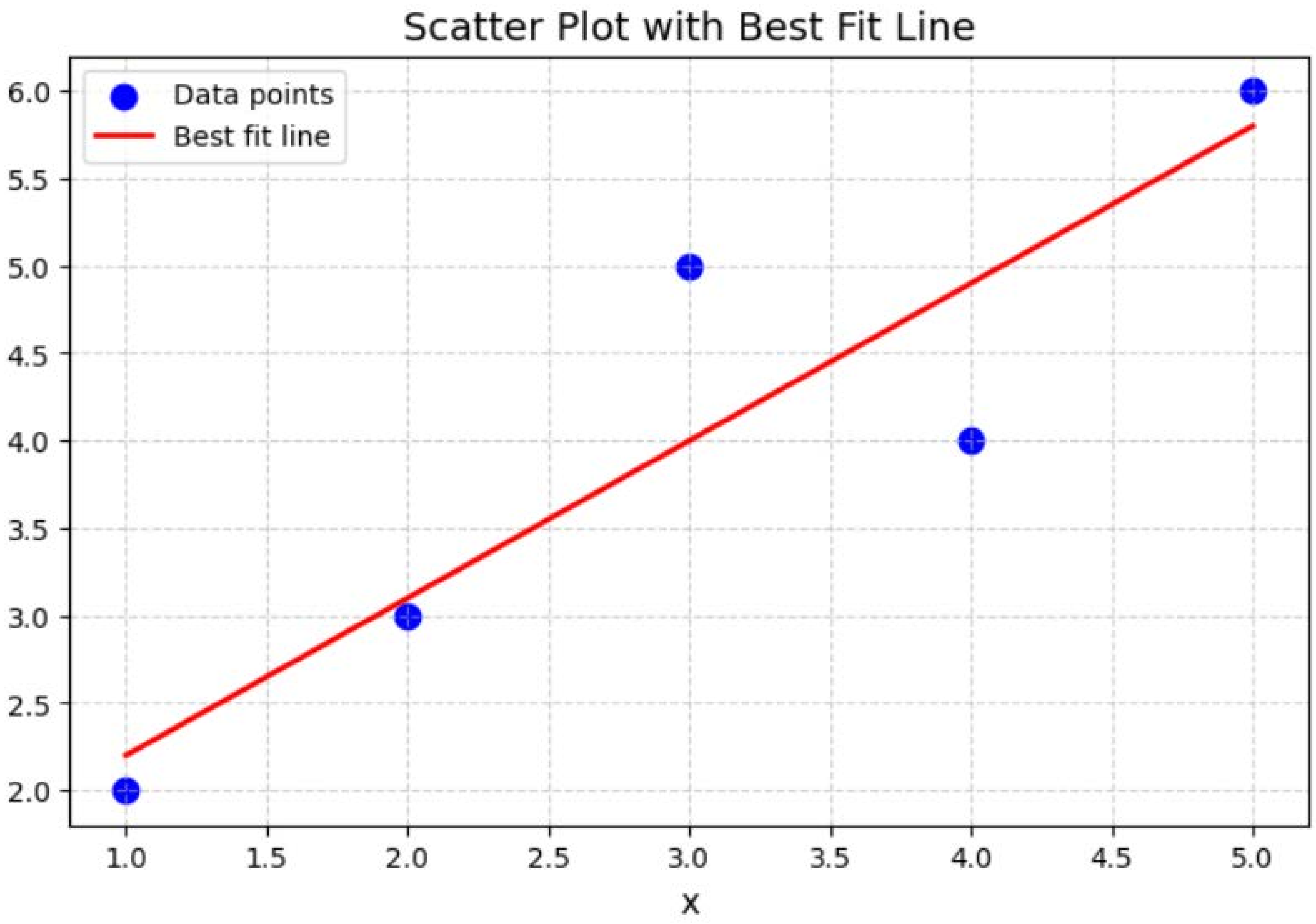


Example

x_i	y_i
1	2
2	3
3	5
4	4
5	6

→From above given data, find the best fit line

→That is, compute the m and c values of the best fit line



Example

i	x_i	y_i	$(x_i)^2$	$x_i y_i$
1	1	2	1	2
2	2	3	4	6
3	3	5	9	15
4	4	4	16	16
5	5	6	25	30
Σ	**15**	**20**	**55**	**69**

Slope m:

$$m = \frac{n \times \sum x_i y_i - (\sum x_i) \times (\sum y_i)}{n \times \sum x_i^2 - (\sum x_i)^2}$$

Intercept c:

$$c = \frac{\sum y_i - m \times \sum x_i}{n}$$

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Slope m:

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Numeric substitution:

$$m = \frac{5 \times 69 - 15 \times 20}{5 \times 55 - 15^2} = \frac{345 - 300}{275 - 225} = \frac{45}{50} = 0.9$$

Intercept c:

$$c = \frac{\sum y_i - m \times \sum x_i}{n}$$

Numeric substitution:

$$c = \frac{20 - 0.9 \times 15}{5} = \frac{6.5}{5} = 1.3$$

Example

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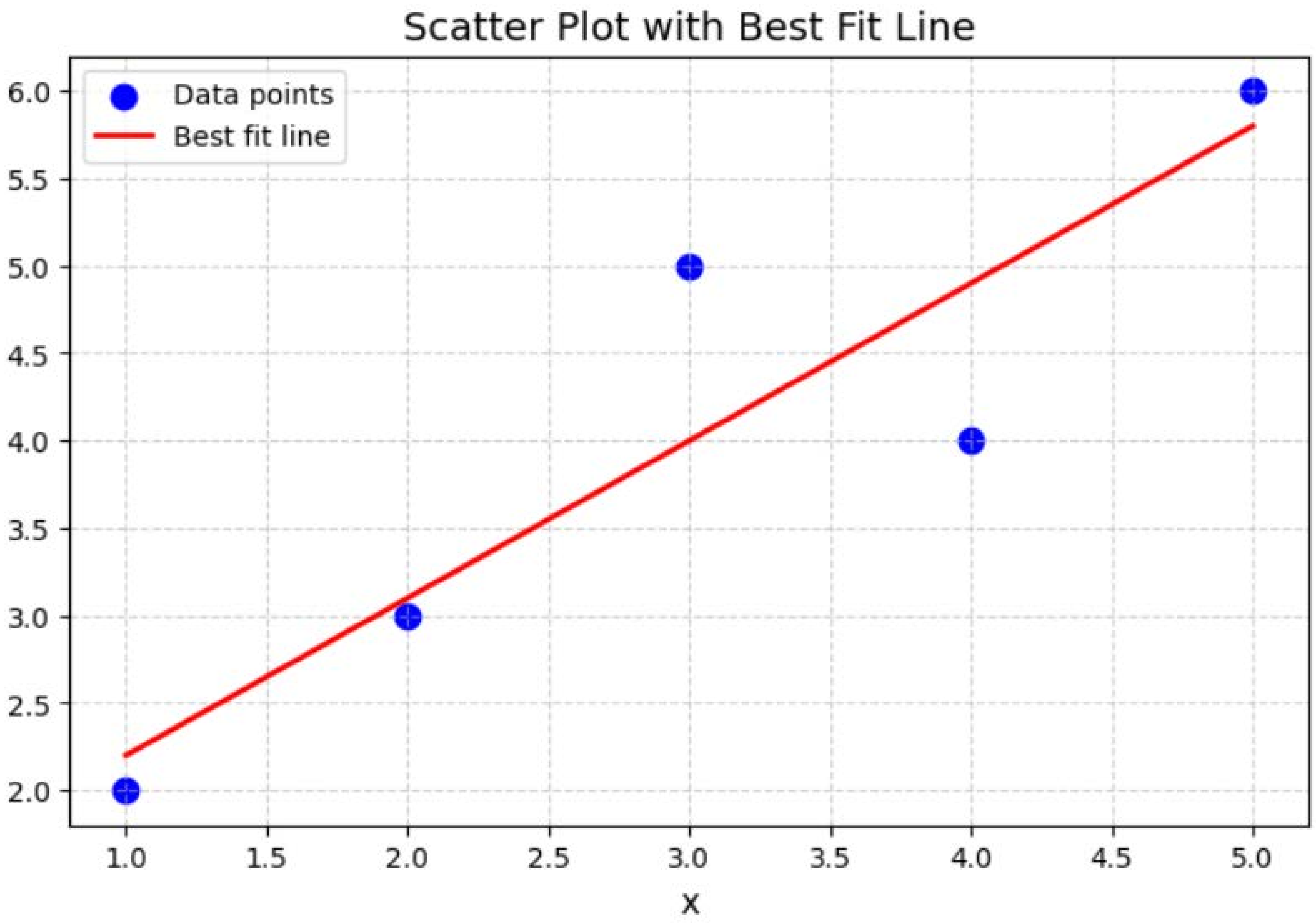
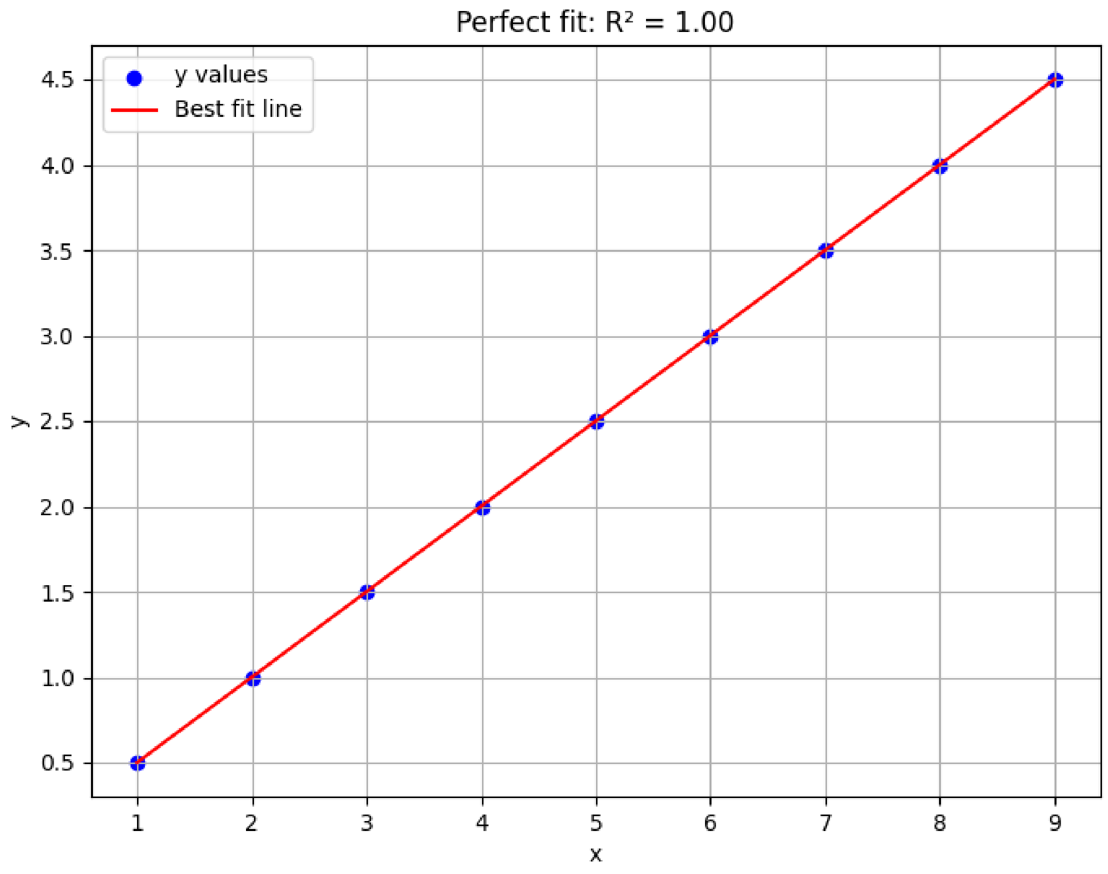
$$y = 1.3 + 0.9x$$



Goodness of fit in linear regression



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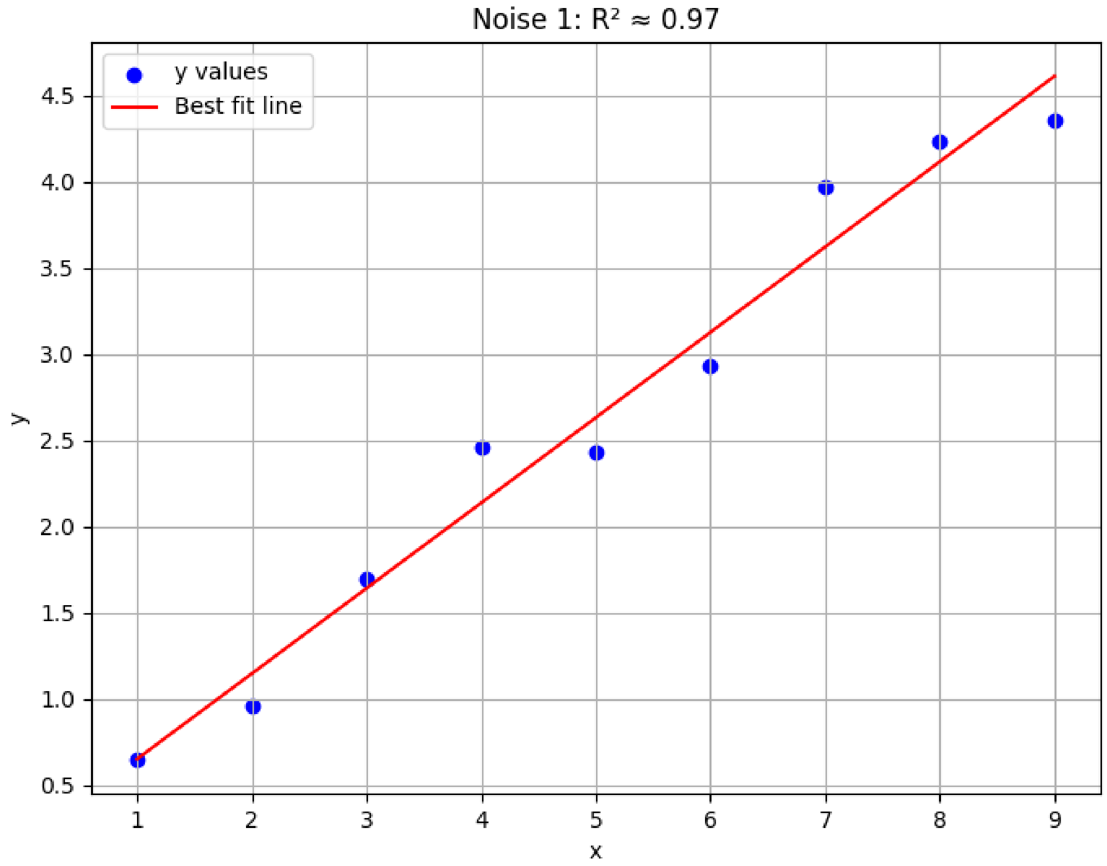

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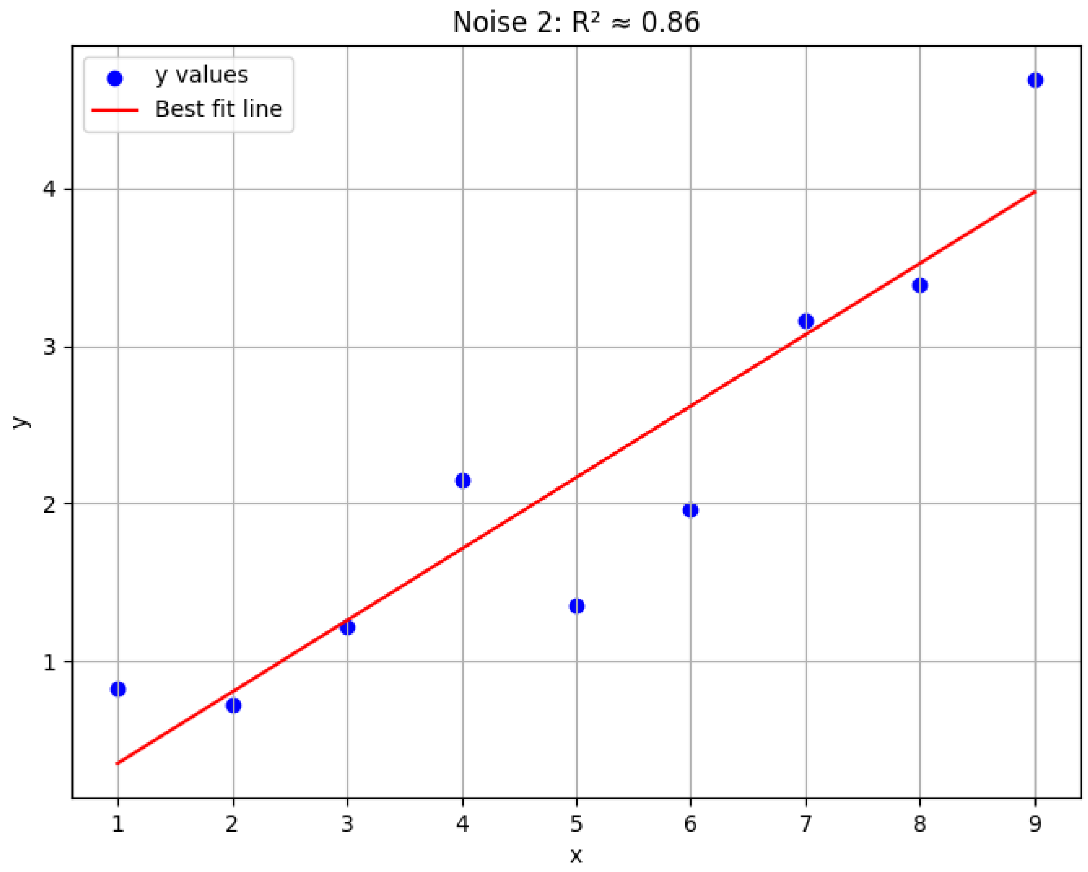

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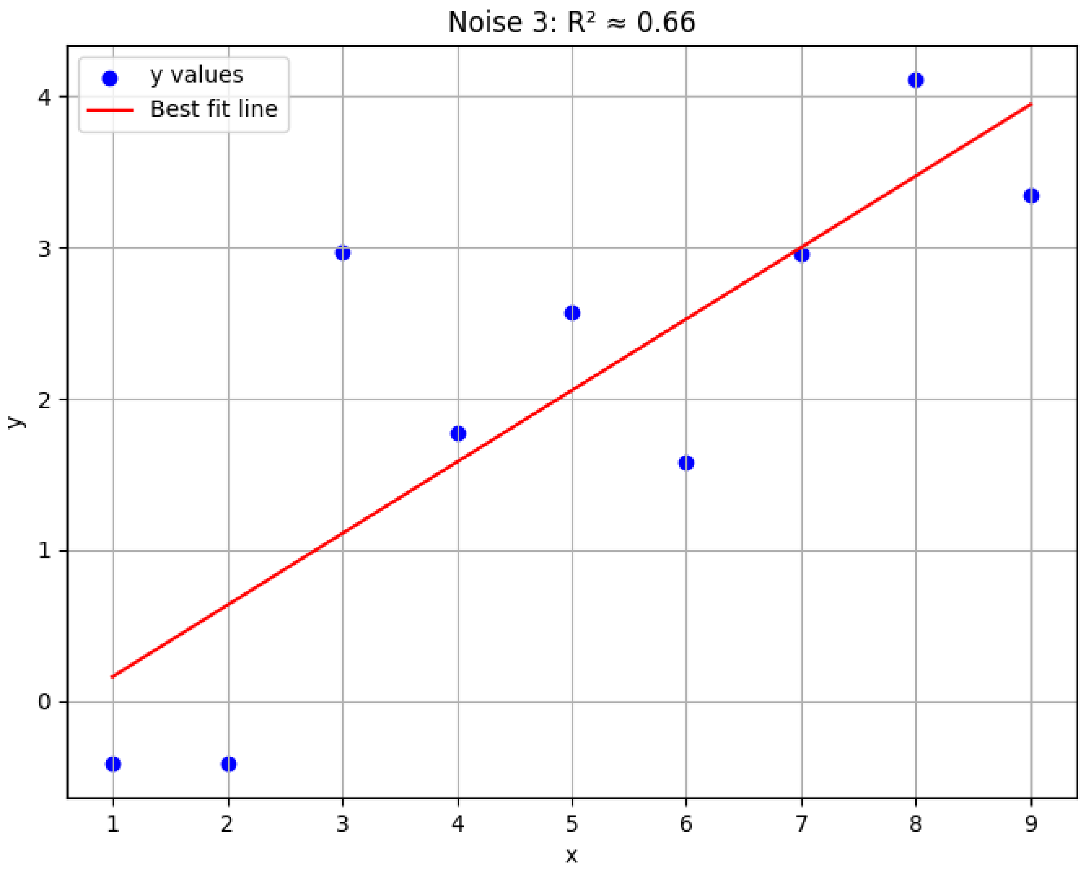

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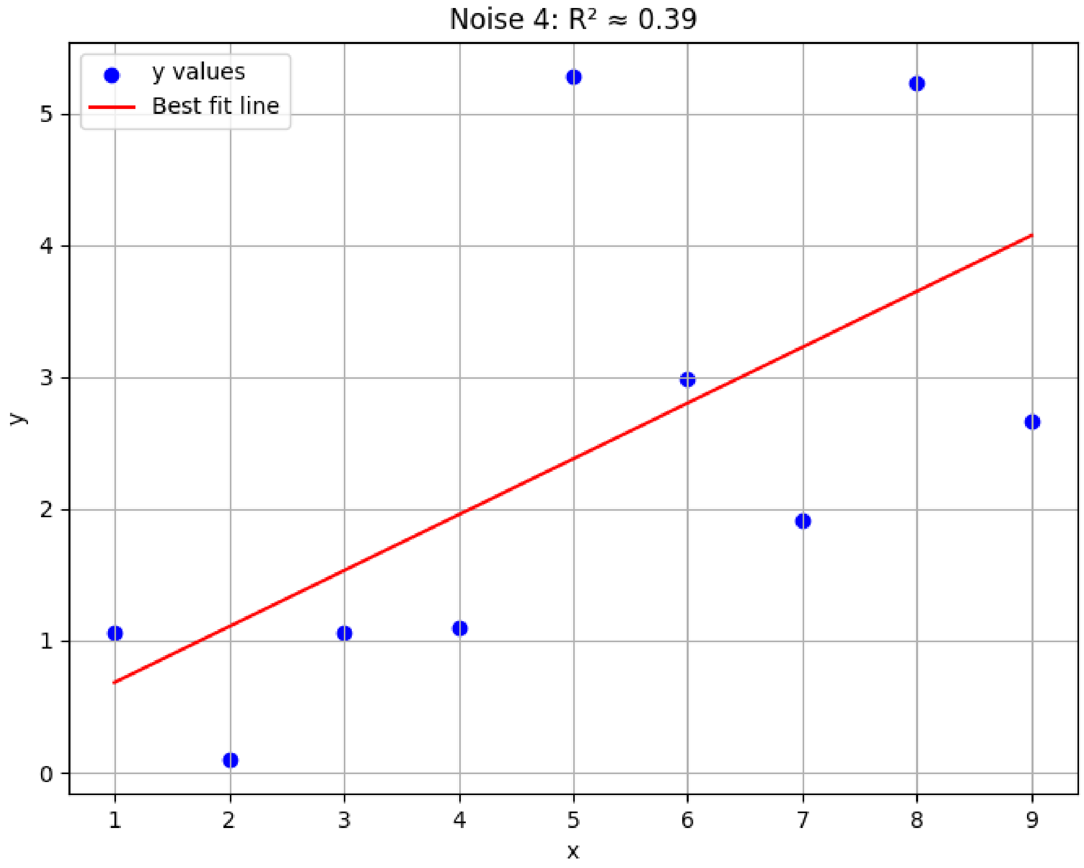


Goodness of fit in linear regression

- **Goodness of fit** tells us **how well the regression line explains the variation in the dependent variable**.
- Essentially:
  - A *good fit* means the *predicted values* are close to the *actual values*.
  - A *poor fit* means the *predicted values* are far from the *actual values*
- Formula

$$R^2 = \frac{\sum (y_p - \bar{y})^2}{\sum (y - \bar{y})^2}$$

y_p = predicted value

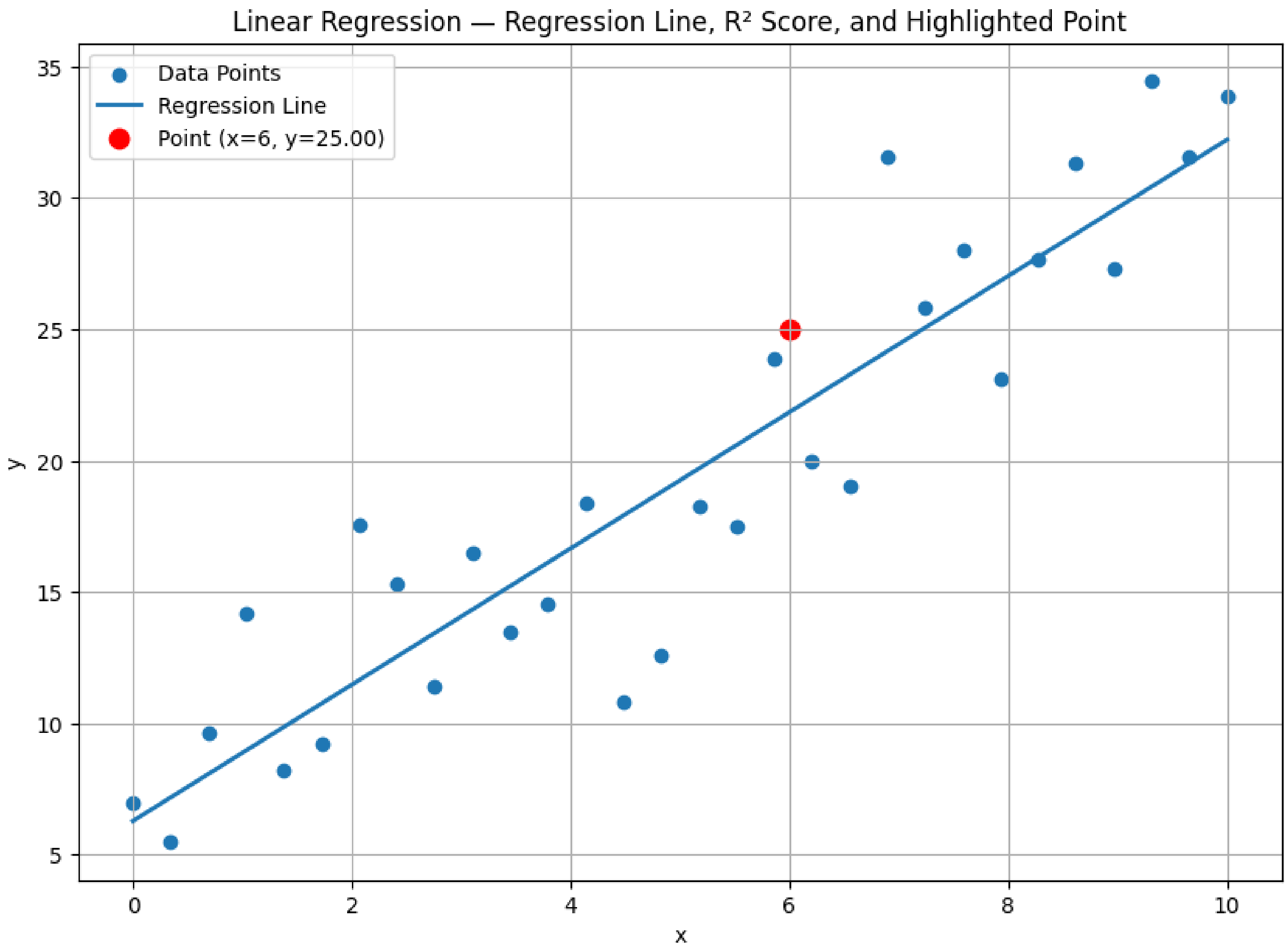
y = actual value

$\bar{y}$ = mean of actual y

Goodness of fit in linear regression

$R^2 = 0.844$

Regression Line Equation:
y = 2.592x + 6.286



Goodness of fit in linear regression

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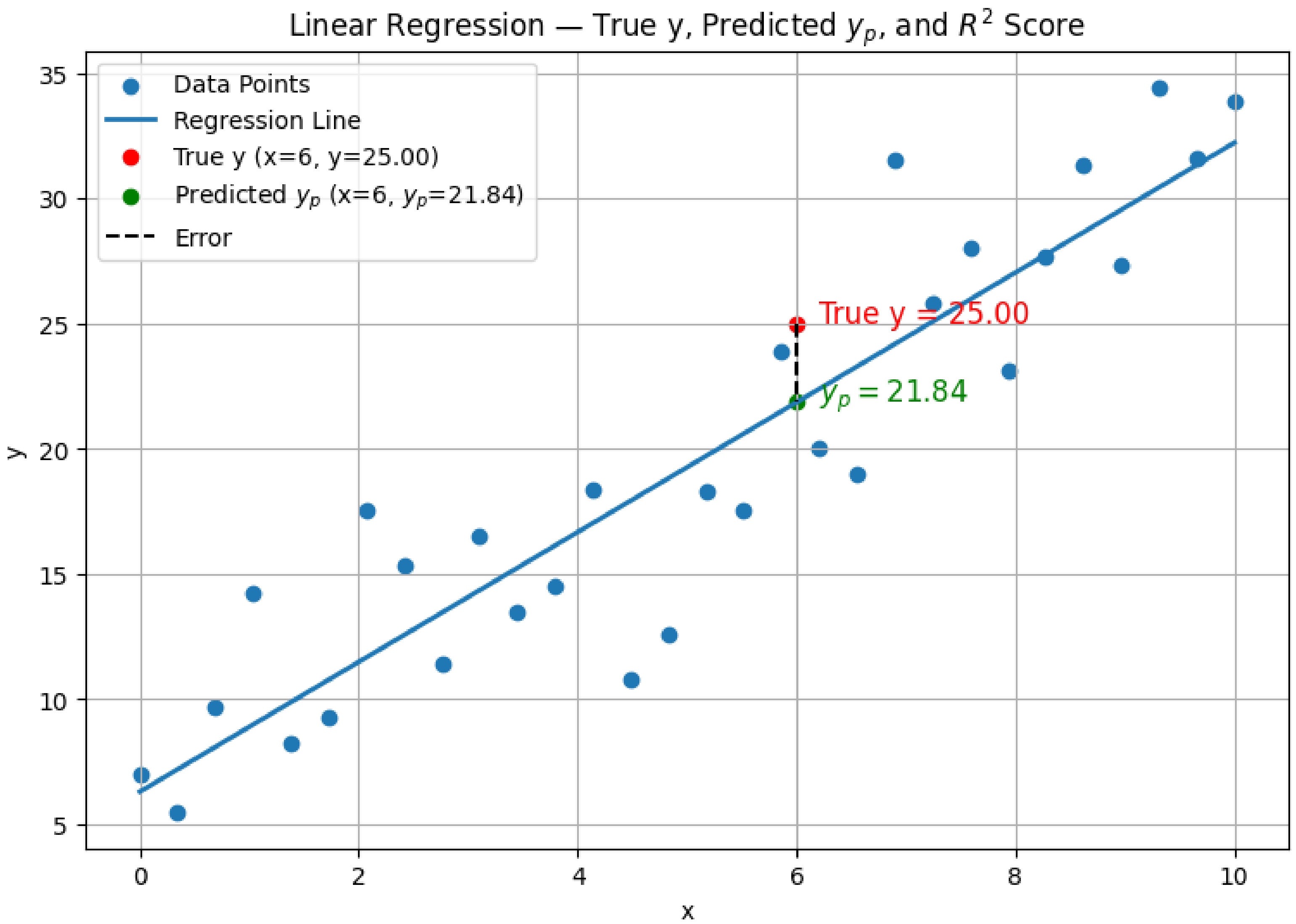

Goodness of fit in linear regression

- **Good Fit**
  - If the regression predicts the data well, y_p will be close to the actual y.
  - In that case,

$$R^2 \approx 0.8\ to\ 1.0$$

Goodness of fit in linear regression

- **Poor Fit**
  - If the regression predicts poorly, y_p will be far from actual y.

$$R^2 \approx 0 \; to \; 0.3$$

Logistic Regression follows....

Thanks